Panskura Banamali College (Autonomous)

B.Sc.(H) 2nd Semester Examination , 2022

Subject: BCA Paper: GE2T

Paper Name : Numerical Computing

F.M. : 40 **Time : 2 Hours**

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1. Answer any four questions: (4x10=40)

a. Why do we bother with Newton Iterations when there are better way to solve things? What is a Taylor Series Expansion? 7.5+7.5

b. Why does the Newton iteration often work better than the secant method for nonlinear functions like f(x,y)?What is the purpose of a Transpose? 8+7

c. Describe the merits of Newton's method of iterations.State the General Newton's- Raphson Method. 8+7

d. State the condition for the convergence of Gauss Seidel iteration method for solving a system of linear equation. State the condition for Convergence of Iteration method . 9+6

e. Calculate the root of $ex - 3x = 0$ in $1 < x < 1.1$ by Iteration method.Give an example of transcendental and algebraic equation . 8+7

f. Find the positive root of x 2-2x -3 = 0 using fixed point iteration method starting with 0.4 as first approximation.Which of the iterative methods is used for solving linear system of equations it converges fast? Why? 5+5+5

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